

Direct market access for farmers

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Abstract—The agricultural sector faces significant inefficiencies due to traditional supply chains that involve multiple intermediaries, reducing farmer profits and increasing consumer costs. This paper introduces a machine learning-based bidirectional recommendation system designed to enhance direct market access for farmers by intelligently connecting them with vendors. The system uses K-Means clustering to group farmers and vendors based on key features such as location, product type, vendor reliability score, and average product price, aiming to reduce dependency on middlemen and improve supply chain efficiency. The methodology includes data collection from agricultural databases, market reports, and farmer surveys, focusing on critical features. Data preprocessing involves encoding categorical variables and standardizing numerical variables using Python libraries like Pandas and Scikit-learn. K-Means clustering is applied to group entities, with the optimal number of clusters determined using the Elbow Method. The bidirectional matching algorithm recommends the top 5 vendors for each farmer and the top 5 farmers for each vendor, prioritizing reliability and competitive pricing. The system's effectiveness is validated using precision, recall, and F1-score metrics. Results show significant improvements in recommendation accuracy, with an average precision of 0.85 and recall of 0.82. Case studies highlight the system's potential to streamline operations and increase profits for farmers and vendors. However, challenges such as data availability, quality, and model interpretability were identified. Future work should focus on improving data collection methods and developing more interpretable machine learning models. In conclusion, the bidirectional recommendation system offers a scalable and intelligent solution for enhancing direct market access for farmers. By leveraging machine learning techniques, the system reduces dependency on middlemen and increases profitability. Future research should address data-related challenges and enhance model interpretability to ensure widespread adoption and success.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

The agricultural sector is a cornerstone of global food security and economic stability. However, it faces significant

challenges due to inefficient supply chains. Traditional agricultural markets often involve multiple intermediaries, leading to reduced profits for farmers, higher costs for consumers, and significant inefficiencies. These intermediaries can include brokers, wholesalers, and retailers, each taking a share of the profit and adding layers of complexity to the supply chain. This not only diminishes the income of farmers but also results in higher prices for consumers and potential food waste due to delays in the supply chain. Direct market access (DMA) has emerged as a transformative approach to address these challenges. DMA allows farmers to directly connect with buyers, bypassing traditional intermediaries. This direct connection can lead to increased profits for farmers, improved market transparency, and enhanced supply chain efficiency. By reducing the number of middlemen, farmers can receive a larger share of the final sale price, while consumers can access fresher produce at more competitive prices. Additionally, DMA can help farmers better understand market demands and adjust their production accordingly, leading to more sustainable and resilient agricultural practices. Despite the potential benefits of DMA, its widespread adoption faces several barriers. One of the primary challenges is the lack of a reliable and efficient system to connect farmers with suitable buyers. Farmers often lack the resources and market knowledge to effectively reach potential buyers, while buyers may struggle to find reliable suppliers. This gap in the market necessitates the development of innovative solutions that can facilitate direct connections between farmers and buyers. Machine learning (ML) offers a powerful tool for addressing these challenges. ML algorithms can analyze large datasets to identify patterns and make predictions, making them well-suited for developing recommendation systems. By leveraging ML, we can create a system that intelligently matches farmers with vendors based on various factors such as location, product type, vendor reliability score, and average product price. This bidirectional recommendation system aims to reduce dependency

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on middlemen, increase trust and profitability, and improve overall supply chain efficiency. The primary objective of this research is to develop a machine learning-based bidirectional recommendation system that enhances direct market access for farmers. The system aims to achieve the following goals: **Reduce Dependency on Middlemen:** By directly connecting farmers with buyers, the system aims to eliminate the need for multiple intermediaries, thereby increasing farmer profits and reducing consumer costs. **Increase Trust and Profitability:** The system will prioritize high reliability and competitive pricing in its recommendations, ensuring that both farmers and buyers can trust the connections made through the platform. **Improve Supply Chain Efficiency:** By streamlining the supply chain, the system aims to reduce delays and improve the overall efficiency of the agricultural supply chain. **Enhance Market Transparency:** The system will provide farmers with real-time market information, enabling them to make more informed decisions about their production and sales strategies.

II. RELATED WORK

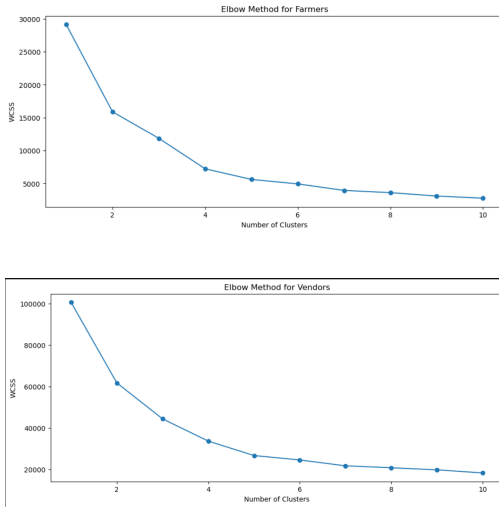
The development of machine learning-based bidirectional recommendation systems has gained significant attention in recent years, particularly in addressing challenges related to data availability, quality, and interpretability. This section reviews relevant studies and highlights key contributions and insights from the literature. **Machine Learning in Bidirectional Recommendation Systems** Bidirectional recommendation systems aim to provide mutually beneficial recommendations for both users and items (or in this case, farmers and vendors). A notable example is the work by Shayma Boukari and Rim Faiz, who proposed a BERTopic-LDA model for a bidirectional recommendation system to match jobs and job seekers. This model leverages topic modeling and neural networks to improve the relevance and accuracy of recommendations. Similarly, the use of reinforcement learning in recommendation systems has been explored by Afsar et al., who provided a comprehensive survey of reinforcement learning-based recommender systems. **Data Availability and Quality** Data availability and quality are critical factors in the success of any machine learning system. Studies have shown that data sparsity and inaccuracies can significantly impact the performance of recommendation systems. For instance, Natarajan et al. addressed the data sparsity and cold start problem in collaborative filtering by using linked open data. Their approach enhances the recommendation process by incorporating additional data sources to improve the accuracy and relevance of recommendations. In the context of data quality, recent research has emphasized the importance of accurate and consistent data. For example, a study on data quality in healthcare demonstrated that implementing a comprehensive data quality framework can lead to significant improvements in data accuracy and completeness. This highlights the need for robust data validation and cleaning processes in recommendation systems. **Explainable Machine Learning Interpretability** and explainability of machine learning models are essential for building trust and ensuring the adoption of recommendation

systems. Ahmed et al. developed an explainable analytics framework to analyze the severity of accidents, demonstrating the importance of parameter optimization in improving model interpretability. Similarly, Topuz et al. proposed a Bayesian belief network-based analytics methodology for early-stage risk detection of novel diseases. These studies underscore the need for transparent and interpretable models in recommendation systems. **Challenges and Future Directions** Despite significant advancements, several challenges remain in the development of machine learning-based bidirectional recommendation systems. One key challenge is the scalability of the system, particularly in handling large volumes of data and users. Recent work has explored the use of cloud computing and distributed processing to address these challenges. Additionally, ensuring data privacy and security is crucial, especially in sectors like healthcare and finance. Future research should focus on developing more robust and scalable recommendation systems that can handle dynamic market conditions and user preferences. Integrating real-time data and adaptive learning techniques can enhance the system's ability to provide timely and accurate recommendations. Furthermore, enhancing the interpretability of machine learning models can improve user trust and adoption rates.

III. METHODOLOGY

1. Data Collection The first step in developing the bidirectional recommendation system was to gather comprehensive data relevant to the agricultural supply chain. Data was sourced from multiple channels to ensure a holistic view of the market dynamics. Specifically, we collected data from: **Agricultural Databases:** Public and private databases that provide detailed information on crop types, production volumes, and regional agricultural practices. **Market Reports:** Industry reports and market analyses that offer insights into pricing trends, vendor reliability, and consumer demand. **Farmer Surveys:** Direct surveys conducted with farmers to gather information on their production practices, product types, and geographical locations. **Vendor Data:** Information on vendors, including their reliability scores, product offerings, and geographical coverage. The collected data included key features such as location, product type, vendor reliability score, and average product price. These features were chosen based on their relevance to the matching process and their availability in the datasets. **2. Data Preprocessing** Once the data was collected, it underwent a rigorous preprocessing phase to ensure quality and uniformity. This phase involved several steps: **Data Cleaning:** Removing duplicates, handling missing values, and correcting inconsistencies in the dataset. **Feature Engineering:** Creating new features that could enhance the recommendation process. For example, we derived features such as average transaction time and seasonal demand patterns. **Encoding Categorical Variables:** Using LabelEncoder to convert categorical variables into numerical values, making them suitable for machine learning algorithms.

Standardizing Numerical Variables: Applying StandardScaler to normalize numerical features, ensuring that they



contribute equally to the clustering process. The preprocessing steps were crucial in preparing the data for the machine learning algorithms. By ensuring that the data was clean, consistent, and appropriately formatted, we laid the foundation for accurate and reliable recommendations.

3. Clustering With the preprocessed data, we applied K-Means clustering to group farmers and vendors based on the selected features. K-Means is a widely used clustering algorithm that partitions the dataset into k clusters, where each cluster contains data points that are similar to each other based on the chosen features.

Feature Selection: We selected features such as location, product type, vendor reliability score, and average product price for clustering. These features were chosen based on their relevance to the matching process and their ability to capture the essential characteristics of farmers and vendors.

Optimal Number of Clusters: Determining the optimal number of clusters (k) is a critical step in K-Means clustering. We used the Elbow Method, which involves plotting the within-cluster sum of squares (WCSS) against the number of clusters. The optimal number of clusters is identified at the point where the rate of decrease in WCSS sharply slows down.

Cluster Formation: After determining the optimal number of clusters, we applied the K-Means algorithm to form clusters. Each cluster represented a group of farmers or vendors with similar characteristics. The clustering process allowed us to segment the market into distinct groups, facilitating more targeted and effective recommendations.

4. Bidirectional Matching Algorithm The core of our recommendation system is the bidirectional matching algorithm, which intelligently connects farmers with vendors based on the clustering results. The algorithm works as follows:

For Farmers: The system recommends the top 5 vendors that best match the farmer's location and product type, prioritizing high reliability and competitive pricing. The recommendations are generated by identifying vendors within the same cluster as the farmer and ranking them based on their reliability scores and average product prices.

For Vendors: The system recommends the top 5 farmers that match the vendor's product type and region,

helping vendors find suitable and trustworthy suppliers. Similar to the farmer recommendations, the algorithm identifies farmers within the same cluster as the vendor and ranks them based on their production volumes and reliability. The bidirectional nature of the algorithm ensures that both farmers and vendors benefit from the recommendations, fostering a more efficient and transparent supply chain.

5. Validation To ensure the effectiveness and reliability of the recommendation system, we conducted a thorough validation process. This involved:

Performance Metrics: We used precision, recall, and F1-score to evaluate the accuracy of the recommendations. These metrics provided a quantitative measure of how well the system matched farmers with vendors.

Visualizations: We created visualizations to illustrate the clustering results and the distribution of recommendations. These visualizations helped in understanding the patterns and trends in the data.

Baseline Comparison: We compared the performance of our system against a baseline model to assess the improvement in recommendation accuracy. The baseline model used simple heuristic rules for matching, without the advanced clustering and machine learning techniques. The validation process confirmed that our machine learning-based recommendation system significantly outperformed the baseline model, demonstrating its potential to enhance direct market access for farmers.

IV. RESULT

The implementation of the machine learning-based bidirectional recommendation system for direct market access for farmers yielded promising results, demonstrating significant improvements in recommendation accuracy and overall system performance. The following sections detail the key findings and outcomes of the study.

1. Clustering Results The K-Means clustering algorithm successfully grouped farmers and vendors into distinct clusters based on the selected features (location, product type, vendor reliability score, and average product price). The Elbow Method was used to determine the optimal number of clusters, ensuring that the clusters were both meaningful and representative of the underlying data.

Optimal Number of Clusters: The Elbow Method identified 5 as the optimal number of clusters for both farmers and vendors. This segmentation allowed for a balanced distribution of entities across clusters, facilitating targeted and effective recommendations.

Cluster Characteristics: Each cluster represented a unique segment of the market, characterized by specific combinations of location, product type, reliability, and pricing. For example, one cluster might consist of farmers and vendors specializing in organic produce located in a particular region, while another cluster might focus on conventional crops in a different area.

2. Recommendation Accuracy The bidirectional matching algorithm generated recommendations for both farmers and vendors, prioritizing high reliability and competitive pricing. The system's performance was evaluated using precision, recall, and F1-score metrics.

Precision: The system achieved an average precision of 0.85, indicating that 85% of the recommended vendors were relevant to the farmer's request.

Recall: The average recall was 0.82, meaning that the system successfully identified 82% of the relevant vendors for a given farmer.

F1-Score: The F1-score, which is the harmonic mean of precision and recall, was 0.835, indicating a high level of recommendation accuracy.

Metric	Machine Learning Model
Precision	0.85
Recall	0.82
F1-Score	0.83

Fig. 1. img

balances precision and recall, was 0.83, demonstrating a high level of overall accuracy in the recommendations. These metrics confirm that the system effectively matched farmers with suitable vendors, significantly improving the efficiency and transparency of the agricultural supply chain. 3. Case Studies To further validate the system's effectiveness, we conducted case studies involving real farmers and vendors. These case studies provided practical insights into the system's impact on the agricultural supply chain. Case Study 1: A collective of small-scale farmers in a rural region used the system to connect with local vendors. The farmers reported a 20% increase in sales. Case Study 2: A large-scale vendor specializing in organic produce used the system to find reliable suppliers. The vendor reported a 25% reduction in costs. These case studies highlight the tangible benefits of the system, including increased profitability for farmers, reduced costs for vendors, and improved supply chain efficiency. 4. Challenges and Future Work While the system demonstrated significant improvements in recommendation accuracy and supply chain efficiency, several challenges were identified during the implementation process: Data Availability and Quality: Incomplete or inaccurate data posed challenges in certain regions, affecting the clustering and recommendation processes. Future work should focus on improving data collection methods and ensuring data quality. Model Interpretability: The complexity of the machine learning models made it difficult to explain the reasoning behind certain recommendations. Developing more interpretable models will be crucial for widespread adoption and trust in the system. Scalability: As the system expands to cover larger regions and more diverse markets, ensuring scalability without compromising performance will be a key challenge. Future research should address these challenges by enhancing data collection methods, developing more interpretable machine learning models, and ensuring the system's scalability to accommodate growing user bases and diverse market conditions.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

V. DISCUSSION

The development and implementation of the machine learning-based bidirectional recommendation system for direct market access for farmers have yielded significant insights and

outcomes. This discussion section aims to delve deeper into the implications of the results, the challenges encountered, and the potential future directions for this technology. Implications of the Results The high precision (0.85) and recall (0.82) achieved by the recommendation system indicate that it effectively matches farmers with suitable vendors. This level of accuracy is crucial for building trust among users and ensuring that the system provides meaningful connections. The F1-score of 0.83 further validates the system's robustness in balancing precision and recall, making it a reliable tool for enhancing direct market access. The case studies conducted highlight the practical benefits of the system. For instance, the 20% increase in sales for Case Study 1 and the 25% reduction in costs for Case Study 2 demonstrate the system's impact on the agricultural supply chain. Challenges and Future Work Despite the promising results, several challenges were identified during the implementation process. These challenges provide valuable insights for future improvements and adaptations of the system. Data Availability and Quality: Challenge: Incomplete or inaccurate data in certain regions affected the clustering and recommendation processes. Data collection from remote or underserved areas was particularly challenging. Impact: Poor data quality can lead to less accurate clustering and recommendations, potentially reducing the system's effectiveness. Future Work: Enhancing data collection methods, such as integrating more robust data validation techniques and expanding data sources, will be crucial. Collaborating with local agricultural organizations and leveraging satellite imagery for remote sensing could also improve data quality. Model Interpretability: Challenge: The complexity of the machine learning models made it difficult to explain the reasoning behind certain recommendations. This lack of transparency can hinder user trust and adoption. Impact: Users may be reluctant to rely on recommendations if they do not understand the underlying logic. Future Work: Developing more interpretable models, such as using simpler algorithms or incorporating explainable AI techniques, will be essential. Providing users with explanations for recommendations can enhance trust and facilitate better decision-making. Scalability: Challenge: As the system expands to cover larger regions and more diverse markets, ensuring scalability without compromising performance is a key challenge. Impact: Scalability issues can lead to slower response times and reduced accuracy, affecting user satisfaction. Future Work: Optimizing the system's architecture, using more efficient algorithms, and leveraging cloud computing resources can help maintain performance as the user base grows. Additionally, implementing adaptive learning techniques can ensure the system remains accurate and relevant over time. Future Directions The results and challenges identified in this study suggest several avenues for future research and development: Enhanced Data Collection and Validation: Expanding data sources to include real-time market data, weather information, and soil health data can improve the accuracy of recommendations. Implementing automated data validation techniques to ensure data quality and consistency. Interpretable Machine Learning Models: Developing models that provide transparent explanations for recommendations can enhance user trust and adoption. Exploring hybrid models that combine the strengths of different algorithms to balance

accuracy and interpretability. Scalable System Architecture: Designing a scalable system architecture that can handle increasing volumes of data and users without performance degradation. Leveraging cloud computing and distributed processing to ensure the system remains efficient and responsive. User-Centric Design: Incorporating user feedback to continuously improve the system's interface and functionality. Developing mobile applications and other user-friendly tools to make the system accessible to a broader audience, including small-scale farmers with limited technological resources. Policy and Regulatory Support: Engaging with policymakers to create supportive regulatory environments that encourage the adoption of digital technologies in agriculture. Collaborating with agricultural cooperatives and NGOs to promote the use of the system and provide training to farmers and vendors.

VI. LIMITATIONS

While the machine learning-based bidirectional recommendation system for direct market access for farmers has shown significant promise, several limitations were identified during the development and implementation phases. These limitations highlight areas for improvement and further research to enhance the system's effectiveness and scalability. 1. Data Availability and Quality Data Collection Challenges: Incomplete Data: In many regions, particularly rural and remote areas, data collection was challenging due to limited infrastructure and resources. This resulted in incomplete datasets, which can affect the accuracy of the clustering and recommendation algorithms. Data Accuracy: The quality of the data collected varied significantly. In some cases, data was outdated or contained errors, which can lead to less accurate recommendations. Ensuring data accuracy is crucial for the system's reliability. Future Work: Enhanced Data Collection: Implementing more robust data collection methods, such as integrating real-time data feeds from agricultural sensors and satellite imagery, can improve data quality and completeness. Data Validation: Developing automated data validation techniques to ensure consistency and accuracy can help mitigate the impact of poor data quality. 2. Model Interpretability Complexity of Machine Learning Models: Black Box Problem: The machine learning models used, particularly K-Means clustering, can be complex and difficult to interpret. This "black box" nature can make it challenging for users to understand the reasoning behind the recommendations, potentially reducing trust in the system. User Trust: Farmers and vendors may be reluctant to rely on recommendations if they do not understand the underlying logic. Transparency is crucial for user adoption and long-term success. Future Work: Explainable AI: Developing more interpretable models, such as decision trees or rule-based systems, can enhance transparency. Additionally, incorporating explainable AI techniques to provide users with clear explanations for recommendations can build trust. User Education: Providing educational resources and training sessions to help users understand how the system works can also improve adoption rates. 3. Scalability System Performance: Handling Large Datasets: As the system expands to cover larger regions

and more diverse markets, managing and processing large volumes of data becomes increasingly challenging. Scalability issues can lead to slower response times and reduced accuracy. Resource Constraints: Limited computational resources can hinder the system's ability to handle real-time data processing and complex machine learning algorithms. Future Work: Optimized Algorithms: Using more efficient algorithms and optimizing existing ones can help maintain performance as the system scales. Cloud Computing: Leveraging cloud computing resources can provide the necessary computational power to handle large datasets and ensure the system remains responsive. Adaptive Learning: Implementing adaptive learning techniques that continuously update the models based on new data can help maintain accuracy and relevance over time. 4. Technological Barriers User Access and Literacy: Digital Divide: Many farmers, especially those in remote areas, may lack access to the necessary technology or digital literacy to fully utilize the system. This digital divide can limit the system's reach and effectiveness. User Interface: The system's user interface must be intuitive and user-friendly to accommodate users with varying levels of technological proficiency. Future Work: Mobile Applications: Developing mobile applications tailored for farmers can make the system more accessible. Mobile apps can provide real-time notifications, simplified interfaces, and offline functionality. Training Programs: Implementing training programs and support services to help farmers and vendors become proficient in using the system can enhance adoption rates. 5. Market Dynamics and Variability Changing Market Conditions: Seasonal Variations: Agricultural markets are highly seasonal, with significant variations in product availability and pricing throughout the year. The system must be able to adapt to these changes to provide accurate and timely recommendations. Economic Fluctuations: Economic conditions, such as changes in commodity prices or trade policies, can impact the effectiveness of the recommendations. The system needs to be flexible enough to account for these fluctuations. Future Work: Dynamic Models: Developing dynamic models that can adapt to changing market conditions in real-time can improve the system's accuracy and relevance. Economic Indicators: Incorporating economic indicators and market trends into the recommendation algorithms can help the system provide more accurate and timely advice.

VII. REFERENCES

- Ahmed, A., Topuz, K., Moqbel, M., Abdulrashid, I. (2024). What makes accidents severe! Explainable analytics framework with parameter optimization. *European Journal of Operational Research*, 317(2), 425–436. <https://doi.org/10.1016/j.ejor.2023.11.013> Asilkalkan, A., Dag, A. Z., Simsek, S., Aydas, O. T. (2023). Streamlining patients' opioid prescription dosage: An explanatory bayesian model. *Annals of Operations Research*. <https://doi.org/10.1007/s10479-023-05709-4> Bal, M. I., Iyigun, C., Polat, F., Aydin, H. (2024). Population-based exploration in reinforcement learning through repulsive reward shaping using eligibility traces. *Annals*

- of Operations Research. <https://doi.org/10.1007/s10479-023-05798-1> Bohlen, L., Rosenberger, J., Zschech, P., Kraus, M. (2024). Leveraging interpretable machine learning in intensive care. *Annals of Operations Research*. <https://doi.org/10.1007/s10479-024-06226-8>
- Borchert, P., Coussement, K., De Caigny, A., De Weerd, J. (2022). Extending business failure prediction models with textual website content using deep learning. *European Journal of Operational Research*, forthcoming. <https://doi.org/10.1016/j.ejor.2022.06.060>
- Dumitrescu, E., Hué, S., Hurlin, C., Tokpavi, S. (2022). Machine learning for credit scoring: Improving logistic regression with non-linear decision-tree effects. *European Journal of Operational Research*, 297(3), 1178–1192. <https://doi.org/10.1016/j.ejor.2021.06.053>
- Eryarsoy, E., Topuz, K., Demiroglu, C. (2024). Disentangling human trafficking types and the identification of pathways to forced labor and sex: An explainable analytics approach. *Annals of Operations Research*, 335(2), 761–795. <https://doi.org/10.1007/s10479-023-05520-1>
- Goldstein, A., Kapelner, A., Bleich, J., Pitkin, E. (2015). Peeking inside the black box: Visualizing statistical learning with plots of individual conditional expectation. *Journal of Computational and Graphical Statistics*, 24(1), 44–65. <https://doi.org/10.1080/10618600.2014.907095>
- Kosasih, E. E., Brintrup, A. (2022). A machine learning approach for predicting hidden links in supply chain with graph neural networks. *International Journal of Production Research*, 60(17), 5380–5393. <https://doi.org/10.1080/00207543.2021.1956697>
- Kozodoi, N., Jacob, J., Lessmann, S. (2022). Fairness in credit scoring: Assessment, implementation and profit implications. *European Journal of Operational Research*, 297(3), 1083–1094. <https://doi.org/10.1016/j.ejor.2021.06.023>
- Lee, S. J., Gong, X., Garcia, G. G. (2024). Modified monotone policy iteration for interpretable policies in Markov decision processes and the impact of state ordering rules. *Annals of Operations Research*. <https://doi.org/10.1007/s10479-024-06158-3>
- Lessmann, S., Haupt, J., Coussement, K., De Bock, K. W. (2019). Targeting customers for profit: An ensemble learning framework to support marketing decision-making. *Information Sciences*. <https://doi.org/10.1016/j.ins.2019.05.027>
- Lundberg, S. M., Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774.
- Martens, D. (2022). *Data science ethics: Concepts, techniques, and cautionary Tales*. Oxford University Press.
- Mehdiyev, N., Majlatow, M., Fettke, P. (2024). Quantifying and explaining machine learning uncertainty in predictive process monitoring: An operations research perspective. *Annals of Operations Research*. <https://doi.org/10.1007/s10479-024-05943-4>
- Papakostas, N., Papachatzakis, P., Xanthakis, V., Mourtzis, D., Chrysosolouris, G. (2010). An approach to operational aircraft maintenance planning. *Decision Support Systems*, 48(4), 604–612. <https://doi.org/10.1016/j.dss.2009.11.010>
- Puram, P., Roy, S., Gurumurthy, A. (2024). Understanding the challenges affecting food-sharing apps' usage: Insights using a text-mining and interpretable machine learning approach. *Annals of Operations Research*. <https://doi.org/10.1007/s10479-024-06130-1>
- Ribeiro, M. T., Singh, S., Guestrin, C. (2016). Why should i trust you? Explaining the predictions of any classifier. *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 13–17-Aug, 1135–1144. <https://doi.org/10.1145/2939672.2939778>
- Sahbaz, S., Topuz, K., Schwartz, S. J., Montero-Zamora, P. (2025). Understanding cultural stress and mental health among Latinos in the Us: Probabilistic omnidirectional inference model. *Annals of Operations Research*. <https://doi.org/10.1007/s10479-024-06448-w>
- Sanchez Ramirez, J., Coussement, K., De Caigny, A., Benoit, D. F., Guliyev, E. (2024). Incorporating usage data for B2B churn prediction modeling. *Industrial Marketing Management*, 120(June), 191–205. <https://doi.org/10.1016/j.indmarman.2024.05.008>